Magnetism in Matter I think I can safely say that nobody understands quantum mechanics. *Richard Feynm*

[width=0.5] /PhyAI/chapter6/$6_1.png$ A loop inside magnetic field $\vec{B}$ A rectangular loop has size $a \times b$, under the net torqu

I define $\vec{n}$ as the normal vector of the loop, and an angle $\theta = (\vec{n}, \vec{B})$, because both torques vectors are in the same direction.

[width=0.5] /PhyAI/chapter6/$6_2.png$ Both torque vectors in pointing out from the plane of paper The scalar τ_{net} value is:

I define a new vector $\vec{\mu}$ quantity, called **magnetic dipole moment**, same direction to normal vector $\vec{n}$, and has magnitude

Rewriting the net torque equation yields:

[Magnetic Moment Definition] The **magnetic moment** (or magnetic dipole moment) is a vector quantity that determines t

Each angle θ (position), it is characterized by a **potential energy value**. I randomly choose 2 angles θ_i and θ_f:

I assign the potential energy value to angle θ:

From here we have a very important result: a magnetic dipole has its lowest energy when its vector $\vec{\mu}$ is lined up with magn

[width=0.666666] /PhyAI/chapter6/$6_3.png$ Highest and lowest potential energy value To flip an arrow (magnetic moment v

[width=1] /PhyAI/chapter6/$6_4.png$ A counterclockwise current inside a diverging magnetic field

[H]

[width=1] /PhyAI/chapter6/$6_5.png$ A clockwise current inside a diverging magnetic field [H]

[width=0.95] /PhyAI/chapter6/$6_6.png$ Two loops system tends to be **repelled away** from the north pole Inside Bohr's ato

[width=0.75] /PhyAI/chapter6/$6_7.png$ The macroscopic view of paramagnetic material An external magnetic field $\vec{B}$ imme

In 1985, Pierre Curie discovered experimentally the relationship:

This proportional relationship seems like very obvious; the higher the temperature, the harder the alignment of magnetic m

[width=0.5] /PhyAI/chapter6/$6_8.png$ Two-state paramagnet If I denote N as "the number of", the multiplicity and entrop

The internal energy is determined by:

I neglect the constant B in the magnetization formula, and redefine the M quantity as:

Before we look for an analytical solution, let's consider the visual representation for these things. We all know how binomia

[width=0.5] /PhyAI/chapter6/$6_9.png$ Relationship between $N_{\updownarrow}, S, U, M$ If our two-state paramagnet starts moving from sta

[width=0.8] /PhyAI/chapter6/$6_10.png$ The graph of (U, S) What can we see here? Recall the definition of temperature (De